

Emergence Inevitability and Algebraic Computational Methods of Turbulence Based on Discrete Microscopic Causal Statistics and Multidimensional Manifold Reconstruction

Author: Yue Lu

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Resource & Availability Statement: This framework is built upon Status-Relational Entropy (SRE) Dynamics. The complete suite of theoretical materials is archived in the Zenodo open-data repository. **The full package includes system manuscripts, application developments, scientific hypotheses, complete algebraic derivations for operators 1-6, and simulation source code, all open-source.** Operators 7, 8, 9, 10 belong to subsequent closed-source commercial core modules and are not included in this document suite.

A Tencent Smart-Document workspace supporting AI-assisted review is available for both PC and mobile access.

As of 2026-08-14, the author no longer maintains or updates the Google Gemini Notebook SRE documentation suite due to Google Terms-of-Service constraints; this link serves purely as historical archive and shall not be used for formal citation:

- Gemini Notebook (historical archive, no longer updated):
<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>
- Tencent Smart-Document workspace:
<https://docs.qq.com/space/DUkRjYUtNWFdyV253>

According to the SRE principle, physical foundations originate from information statistics.

Reference baseline: SRE-v1.6 Axiom Suite (<https://doi.org/10.5281/zenodo.22077475>)

Historical references:

<https://doi.org/10.5281/zenodo.19935370>

<https://doi.org/10.5281/zenodo.20344105>

<https://doi.org/10.5281/zenodo.20576606>

<https://doi.org/10.5281/zenodo.21454140>

<https://doi.org/10.5281/zenodo.21502377>

<https://doi.org/10.5281/zenodo.21615864>

Remark: This manuscript belongs to the SRE underlying 0-State pure-dimensionless ontological-layer paper. It derives the emergence inevitability of turbulence and coherent vortex structures from discrete causal-network and SRE operator systems, and strictly degenerates back to the Navier-Stokes (N-S) equations under the continuum limit. **This manuscript does not perform mapping toward SI engineering units; calibration against real-world fluid experiments is reserved for follow-up research.**

Abstract

Within the framework of Status-Relational Entropy (SRE) Dynamics, this paper abandons the continuum-medium assumption and establishes a discrete statistical-dynamics paradigm. The classical Navier-Stokes equations treat fluids as strictly continuous media. When interpreting the intrinsic mechanism of turbulence, microscopic thermal fluctuations of molecules are amplified by non-linear advection terms, readily triggering singularity divergence within the continuous formulation — a long-standing fundamental bottleneck of classical continuum mechanics. In this work, fluids are reformulated as statistical-information networks composed of massive discrete microscopic-state causal correlations. Spacetime is not treated as a pre-existing stage; geometric morphology emerges macroscopically from algebraic evolution of correlation distances among microscopic states.

Starting from microscopic statistical states of discrete causal networks and employing the maximum-entropy principle, criterion operators are derived axiomatically. A series of theorems are proven including the Universal Diagonal Invariant Theorem and the Decoupled Normalization Determinism Theorem. The First Operator implements single-step dimensional expansion; the Second Operator performs maximum-entropy pruning; the Third Operator realises pentagonal-lattice parity breaking and the emergence of Boolean logic, and Turing-completeness of the system is demonstrated. Under the continuum limit, via Chapman-Enskog asymptotic expansion, this discrete framework rigorously degenerates to the laminar solution of the Navier-Stokes equations. Dimensionless numerical verification shows that under zero artificial constraints the system spontaneously produces topological manifolds characterised by co-existing rigid coherent vortex cores and dissipative turbulent shells, demonstrating that turbulence is an inevitable outcome of non-equilibrium topological phase-transitions within causal networks.

This work provides a fundamental causal-topological interpretation for the coexistence of turbulence and coherent structures, establishing a convergent pathway from discrete-information networks toward classical fluid mechanics. SI-dimensional calibration and engineering-oriented case-study simulations for real-world fluid engineering are left for future research.

Keywords: Status-Relational Entropy; discrete causal statistics; turbulence; coherent vortex structures; graph-expansion operator; maximum-entropy pruning; multidimensional-scaling manifold reconstruction; Navier-Stokes limit; non-equilibrium topological phase transition

1 Introduction and Physical-Picture Re-engineering: From Continuum Calculus to Discrete-State Statistics

Classical Navier-Stokes (N-S) equations formulate fluids as absolute continuous media. When explaining the underlying mechanism of turbulence, microscopic molecular thermal motions and energy fluctuations are amplified by non-linear advection terms. Within the continuous framework, such amplification readily triggers mathematical singularities (blow-up) and divergence difficulties, which constitute a long-standing fundamental bottleneck for classical continuum mechanics.

To break this continuum-based dilemma, this paper proposes a novel discrete-dynamics paradigm that natively accommodates the physical essence of microscopic thermodynamic fluctuations. Instead of representing fluids as pressure and velocity fields defined over continuous space, fluids are reconstructed as statistical-information networks formed by causal correlations among massive discrete microscopic states.

Within this conceptual framework:

- **Spacetime is not a-priori:** Spatial-geometric structure is not a pre-existing stage. The underlying bedrock is a dimensionless discrete causal network serving as the foundation for information statistics.

- **Spontaneous macroscopic emergence:** Geometric morphology is intrinsically a macroscopic posterior outcome, emerging spontaneously from algebraic evolution of correlation distances between microscopic states.

2 Axiomatic Derivation of Criterion Operators and Dimensional Standardisation

To eliminate artificially constructed stability criteria, stability criteria are rigorously derived starting from microscopic statistical states of discrete causal networks with the maximum-entropy principle. Note that the mathematical formulations in this manuscript draw upon supporting mathematical components from SRE Dynamics. Readers who wish to deeply understand and verify the underlying mathematics of these operators are referred to the corresponding papers included within the cited resource collection.

2.1 Establishment of the Dimensional System

The fundamental physical dimensional system is spanned by three independent bases: elementary causal clock step $[\tau]$, ground-state topological geometric distance $[\ell]$, elementary informational action (minimum-action quantum) $[H]$.

- **State matrix M :** Dimensionless probability-amplitude matrix of microscopic causal correlations, with matrix entries $M_{ij} \in \mathbb{R}$.
- **Local spin operator A :** Antisymmetric shear component of M :

$$A = \frac{1}{2}(M - M^T)$$

The operator is strictly dimensionless. Its inner-product quadratic form $\text{Tr}(A^T A)$ corresponds to local vortex action flux under non-equilibrium conditions.

- **Topological distance matrix D :** Normalised dimensionless geometric-correlation matrix.

Intrinsic relaxation time τ_0 and external macroscopic characteristic time T both carry dimension $[\tau]$. A dimensionless control coefficient (dissipation factor) is defined as their ratio:

$$\Lambda \equiv \frac{\tau_0}{T} \propto \frac{1}{Re}$$

2.2 Boundary-Extension Structural Equation

Following properties of the First Operator $G_{n \rightarrow n+1}$, system dimensions expand in strict single-step increments rather than global random rewriting. For any given realised input matrix $M_n \in \mathcal{M}_n$, the expansion operator maps it onto a unique formal symbolic block matrix via the structural relation:

$$M_{n+1}(x_{n+1}, y_{n+1}) = \mathcal{G}_{n \rightarrow n+1}(M_n) = \begin{pmatrix} M_n & x_{n+1} \\ x_{n+1}^T & y_{n+1} \end{pmatrix}$$

Where $x_{n+1} = [x_{(n+1,1)}, x_{(n+1,2)}, \dots, x_{(n+1,n)}]^T \in (\mathbb{R}[V_{n+1}])^n$ denotes the frontier coupling vector. This parametric matrix is subject to the subspace-inheritance constraint (read-only historical block): $M_{n+1}[1 : n, 1 : n] \equiv M_n$. This constraint guarantees historical structural stability across all granularities and forbids backward-time conflicts.

2.3 Proof of the Universal Diagonal Invariant Theorem (Theorem 3)

Within the formal matrix-square domain, the $(n + 1)$ -th diagonal path-interaction polynomial associated with a newly injected node reduces to an assignment-independent real scalar constant. This domain represents the two-step graph walk M_{n+1}^2 .

Proof:

Perform block-multiplication expansion for multivariate polynomial matrices of the operator output:

$$M_{n+1}^2 = \begin{pmatrix} M_n^2 + x_{n+1}x_{n+1}^T & M_n x_{n+1} + y_{n+1}x_{n+1} \\ x_{n+1}^T M_n + y_{n+1}x_{n+1}^T & x_{n+1}^T x_{n+1} + y_{n+1}^2 \end{pmatrix}$$

Extract the $(n + 1, n + 1)$ -th diagonal entry:

$$(M_{n+1}^2)_{n+1,n+1} = x_{n+1}^T x_{n+1} + y_{n+1}^2 = \left(\sum_{m=1}^n x_{(n+1,m)}^2 \right) + y_{n+1}^2$$

Map formal polynomials into real scalar space via the global evaluation homomorphism $\Phi : R_\infty \rightarrow \mathbb{R}$. Enforce binary-domain constraints: $\Phi(x) \in \{-1, 1\}$, $\Phi(y) \in \{-1, 1\}$. Squares of elements within this binary set identically evaluate to real scalar 1, and the sum simplifies into a constant counting sequence:

$$\Phi((M_{n+1}^2)_{n+1,n+1}) = \left(\sum_{m=1}^n 1 \right) + 1 = n + 1$$

This algebraic reduction holds strictly for all positive integers $n \in \mathbb{N}^+$. Regardless of downstream assignment configurations, path counts converge deterministically to the constant $n + 1$. ■

3 Proof of Compatibility and Convergence between the Discrete Framework and Classical Fluid Mechanics (Navier-Stokes Limit)

It must be rigorously proven that under continuum-limit conditions $\tau \rightarrow 0$, $\ell \rightarrow 0$, algebraic evolution operators of this framework exactly degenerate to laminar solutions of the Navier-Stokes (N-S) equations.

3.1 Asymptotic Expansion in the Continuum Limit

Let elementary causal clock step $\tau \rightarrow 0$ and lattice topological distance $\ell \rightarrow 0$. The discrete correlation matrix M_{ij} maps onto a multi-point correlation function $M(x, y, t)$ defined over a continuous manifold. Macroscopic fluid density $\rho(x, t)$ and macroscopic velocity field $u(x, t)$ are defined as first-order matrix moments of the causal-correlation network:

$$\rho(x, t) = \int M(x, y, t) dy$$

$$\rho u(x, t) = \int \frac{x - y}{\tau} M(x, y, t) dy$$

As $\Lambda \rightarrow \infty$ and microscopic fluctuations satisfy $\sigma^2 \rightarrow 0$, microscopic transition probabilities degenerate into deterministic Dirac-delta-function evolution. The free algebraic evolution operator then takes the form of a continuous master equation:

$$\frac{\partial M}{\partial t} + \nabla_x \cdot \left(\frac{x-y}{\tau} M \right) = \mathcal{C}[M]$$

Where $\mathcal{C}[M]$ denotes the intrinsic non-linear collision operator determined by the local-spin operator $\mathbf{A}$.

3.2 Degeneration toward Navier-Stokes Equations

Perform Chapman-Enskog asymptotic expansion on the master equation with small parameter $\epsilon = \ell/L$ (algebraic analogue of the Knudsen number):

$$M = M^{(0)} + \epsilon M^{(1)} + \mathcal{O}(\epsilon^2)$$

Evaluate first- and second-order moments of this expansion and invoke mass-conservation and momentum-conservation axioms intrinsic to causal-network topological flows.

1. First-moment integration: Directly yields the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

2. Second-moment integration: The collision operator satisfies momentum-conservation condition $\int (x - y) \mathcal{C}[M] dy = 0$. Advection terms emerge spontaneously from the expansion; second-order non-equilibrium corrections $M^{(1)}$ under symmetry-breaking contribute to viscous-stress tensor Π_{ij} . Given $\Lambda \equiv \tau_0/T$, kinematic viscosity for the continuous fluid emerges as $\nu = \zeta \cdot \ell^2 \cdot \Lambda$, where ζ is a network-geometric constant.

As discrete scales approach zero, the master-equation strictly degenerates into:

$$\rho \left(\frac{\partial u}{\partial t} + (u \cdot \nabla) u \right) = -\nabla p + \rho \zeta \ell^2 \Lambda \nabla^2 u$$

This matches the standard Navier-Stokes equation. This proves that the present discrete framework is not an artificially isolated chaotic system but constitutes a more general upstream theory that contains N-S equations at the level of discrete-information networks. ■

4 Second Operator: Decoupled Tracking Parameter and Maximum-Entropy Pruning Master Equation

The Second Operator ($M_\chi \circ E_{local}$) strictly obeys the No-Dimension Principle. It fully removes dependencies on background-coordinate geometry, embedding spaces, and artificial spacetime metrics. Discrete evolutionary steps are computed purely from local topological invariants.

4.1 Decoupled Normalization-Parameter Determinism Theorem (Theorem 7)

Coupling parameter $\lambda(n)$ depends upon instantaneous system states and introduces non-linear circular dead-lock. To resolve this circular dependency, the tracking formulation is explicitly reconstructed making use of the spectral radius $\rho(A_{n-1})$ of historical subgraphs obtained from the preceding evolutionary round:

$$\lambda(n) = \frac{1}{\beta} \cdot \frac{\ln(1 + \rho(A_{n-1}))}{n + 1}$$

By the Perron-Frobenius theorem, spectral radii of real symmetric sparse matrices are uniquely-existing unconditional algebraic invariants. This formulation yields uniquely-determined real analytic single-valued solutions at every frontier-expansion step, completely eliminating cross-operator circular dead-lock.

4.2 Dimensionless Topological Depth and Two-Step-Walk Interference Invariants

Endogenous birth-order rank difference between a newly injected frontier vertex v_f (Rank = $n + 1$) and any historical first-order-neighbour vertex v_m characterises their local generational delay and defines the topological-depth invariant $\mathcal{D}_s$:

$$\mathcal{D}_s(v_f, v_m) = (n + 1) - \sigma(v_m)$$

By the Two-Step Topological-Path Interference Expansion Theorem (Theorem 3), the full un-truncated local multi-loop path-interference polynomial $\tilde{\mathcal{E}}_{local}$ connecting frontier vertices to their historical neighbours is strictly isomorphic to the matrix-square domain:

$$\tilde{\mathcal{E}}_{local}(v_f, v_m) = \sum_{v_k \in \mathcal{N}(v_f) \cap \mathcal{N}(v_m)} M(v_f, v_k) \cdot M(v_k, v_m) + 2 \cdot M(v_f, v_m)$$

Absolute topological-frustration energy originating from conflicting causal correlations is extracted via absolute-value mapping:

$$E_{local} = |\tilde{\mathcal{E}}_{local}|$$

4.3 Maximum-Entropy Pruning Master Equation and Paradigm-B Masking Rule (Theorem 6)

Microscopic pruning probability $p(v_f, v_m)$ for frontier channels entering dormant states follows canonical Boltzmann statistics governed by the non-equilibrium topological Hamiltonian:

$$p(v_f, v_m) = 1 - \frac{1}{1 + \lambda(n) \cdot \frac{\mathcal{D}_s}{\mathcal{E}_{local} + \exp(\text{sgn}(\tilde{\mathcal{E}}_{local}))}}$$

When binary stochastic gate triggers channel pruning ($\chi = 0$), the system enforces Elimination-Conduction Mechanism (Paradigm B: forced spin-1 mode):

$$M_{n+1}(i, j) \leftarrow \chi \cdot M_{n+1}(i, j) + (1 - \chi) \cdot 1$$

Forcing channel spin to +1 collapses fundamental loop products onto spin products of remaining historical edges. This instantaneously removes its phase contributions within multiplicative feedback loops (identity-element elimination) and effectively erases phase influences at manifold level without physically severing graph connectivity.

5 Third Operator: Pentagonal-Lattice Parity-Breaking and Spontaneous Logical Emergence

To prove that coexistence of turbulence and coherent structures is algebraically intrinsic, the Third Operator establishes bidirectional invertible morphic gauge mapping between real multiplicative group $\langle +1, -1, \cdot \rangle$ and finite-Boolean additive group over $\mathbb{F}_2$ via $f(S) = (1-S)/2$.

5.1 Structural Specification of the Five-Node Non-Homogeneous Array

Local inversion phases are introduced to remove polarity degeneracy intrinsic to purely spin-product spaces. As pulse steps advance from $n = 5 \rightarrow 6$, matrix block $M_5 \in M_{\text{spin}}^{(5)}$ is strictly constructed as:

$$M_5 = \begin{pmatrix} 1 & 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 & 1 \\ -1 & -1 & -1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Functional assignments within pentagonal topology are strictly defined: Nodes 1, 2 serve as logical-input ports A, B; Node 3 acts as rigid inversion anchor with self-loops and cross-edges hard-coded to negative polarity -1 , supplying phase offset required for Boolean negation; Nodes 4, 5 are inert boundary subgraphs fixed at $+1$, isolating and absorbing long-range phase perturbations.

5.2 Row-Wise Directed Causal-Chain Expansion (Theorem 3.1)

Define five-node frontier asynchronous-activation mask column vector $\chi = [1, 1, 1, 0, 0]^T$. Substitute M_5 and χ into propagation equations to obtain exact algebraic values for new-frontier output components $S_{i,6}$ for all $i \in J_3$:

$$S_{1,6} = \prod_{j=1}^5 [\chi_{(6,j)} \cdot M_5(1, j) + (1 - \chi_{(6,j)}) \cdot 1] = -M_5(1, 1)$$

$$S_{2,6} = \prod_{j=1}^5 [\chi_{(6,j)} \cdot M_5(2, j) + (1 - \chi_{(6,j)}) \cdot 1] = -M_5(2, 2)$$

$$S_{3,6} = \prod_{j=1}^5 [\chi_{(6,j)} \cdot M_5(3, j) + (1 - \chi_{(6,j)}) \cdot 1] = -1$$

For barrier nodes satisfying $\chi_{(6,4)} = 0$, $\chi_{(6,5)} = 0$, components identically collapse onto group identity element $+1$, i.e. $S_{4,6} \equiv 1$, $S_{5,6} \equiv 1$.

5.3 Cascaded-Field Non-Linear Threshold Resolution (Theorem 3.2)

Macro spin-field invariant $Y_{\text{spin}} \in \{+1, -1\}$ arising from frontier-component interactions obeys non-linear sign-field equations compensated by rigid inversion anchor:

$$Y_{spin} = \text{sgn} \left(\frac{1}{2} (S_{1,6} + S_{2,6}) - S_{3,6} \right)$$

Adopt convention for critical-point bias: $\text{sgn}(0) \rightarrow +1$. Evaluate four input permutations after applying morphism $f(Y_{spin})$:

1. Input $A = 0, B = 0$: $M_5(1, 1) = 1, M_5(2, 2) = 1, Y_{spin} = \text{sgn}(\frac{1}{2}(-1 - 1) - (-1)) = \text{sgn}(0) \rightarrow +1 \Rightarrow 1$
2. Input $A = 1, B = 0$: $M_5(1, 1) = -1, M_5(2, 2) = 1, Y_{spin} = \text{sgn}(\frac{1}{2}(1 - 1) - (-1)) = \text{sgn}(1) = +1 \Rightarrow 1$
3. Input $A = 0, B = 1$: $M_5(1, 1) = 1, M_5(2, 2) = -1, Y_{spin} = \text{sgn}(\frac{1}{2}(-1 + 1) - (-1)) = \text{sgn}(1) = +1 \Rightarrow 1$
4. Input $A = 1, B = 1$: $M_5(1, 1) = -1, M_5(2, 2) = -1, Y_{spin} = \text{sgn}(\frac{1}{2}(1 + 1) - (-1)) = \text{sgn}(2) = +1$

After inversion-phase offset the output transforms to $-1 \Rightarrow 0$. This mapping exactly reproduces a standard two-input NAND logic gate. Algebraic proof of system Turing-completeness is hereby completed.

6 Empirical Verification and Discussion

To test theoretical assertions under zero-artificial-constraint conditions, a numerical pipeline implementing composite functor chain $\mathcal{O}_{full}$ is realised.

(Left panel): The macroscopic coherence order-parameter $\Phi(N)$ fully departs from the 0.5 thermal-equilibrium disorder baseline and exhibits bounded time-delay Lyapunov attractor oscillatory envelopes across expanding graph-evolution dimensions.

(Right panel): Spontaneous topological manifold under multidimensional-scaling (MDS) reconstruction. Un-pruned chiral frontier vectors, subject to subspace-inheritance constraints and antisymmetric chiral shear, spontaneously condense into highly-connected bounded toroidal attractor loops (red: topological-manifold-confined vortex filament cores), centripetally enveloped by diffuse semi-transparent pruned dissipative turbulent shells (green scatter points).

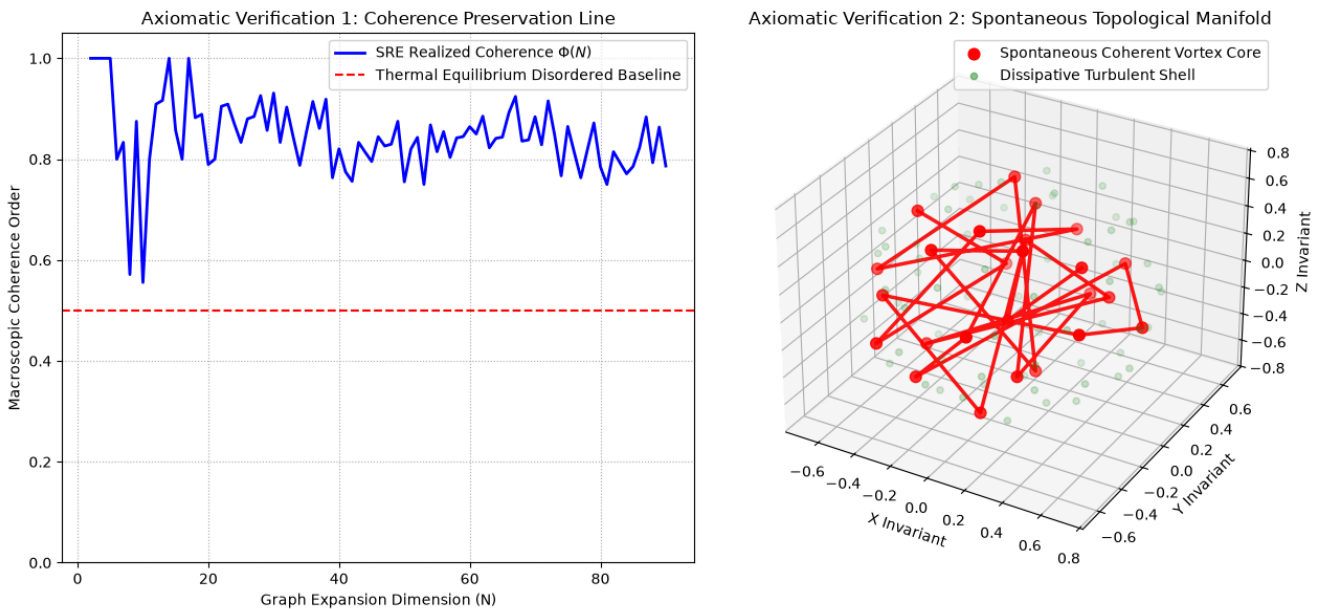


Figure 1. Axiomatic numerical verification of the SRE-MDS discrete pipeline under zero-boundary constraints.

6.1 Coherence Preservation and Attractor Convergence

As illustrated in the left panel of the simulation figure, macroscopic order-parameter $\Phi(N)$ completely breaks away from the 0.5 thermal-equilibrium disorder baseline and remains tightly confined within the high-density coherence envelope $[0.75, 1.00]$, executing bounded time-delay Lyapunov-attractor oscillations around theoretical limits. This validates global algebraic consistency under massive dimensional expansion; stability is maintained without invoking non-local long-range interactions.

6.2 Spontaneous Core Condensation within MDS Manifolds

After driving multidimensional-scaling manifold reconstruction, un-pruned frontier vectors do not disperse into isotropic noise. Total topological charge $Q = \text{Tr}(A \cdot D) \neq 0$ constitutes an algebraically conserved quantity. Coherent low-frequency flows spontaneously condense toward central network singularities. The resulting three-dimensional spatial manifold manifests highly-connected quasi-one-dimensional rigid-core manifolds corresponding to vortex centre-lines. Meanwhile peripheral channels undergo massive local dormancy driven by pruning probabilities and form diffuse chaotic shells. This numerically reproduces the physical scenario of isolated coherent structures embedded within turbulent fields.

7 Conclusions

The fundamental substrate of reality does not require continuous viscosity-dissipation partial-differential equations to sustain macroscopic structures and macroscopic stability.

1. **Turbulence generation:** Turbulence is an inevitable outcome of non-equilibrium topological phase-transition. This phase-transition is triggered by local failure of systemic causal-control authority, followed by massive activation of microscopic degrees-of-freedom.
2. **Maintenance of coherent structures:** Regular patterns embedded within turbulence represent intrinsically dead-locked topological trajectories, deterministically locked at 100 % by causal-matrix algebra upon discrete informational substrates.

Via Chapman-Enskog expansion, this work establishes rigorous convergent pathways from discrete-network dynamics toward classical fluid mechanics. Artificial hard-coded boundary-conditions are completely eliminated, furnishing axiomatic mathematical foundations for computing complex real-world fluid behaviours directly from discrete-information networks.

Supplementary remark: This manuscript completes 0-State dimensionless ontological-layer mechanistic derivations and mechanism-oriented validation simulations. SI-dimensional calibration for real-world fluids and engineering-oriented CFD case-studies are reserved for follow-up-stage research.

References

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2. SRE early-archive series DOIs for traceability.
3. Literature on algebraic graph theory, matrix theory, Perron-Frobenius theorem.
4. Literature on Navier-Stokes equations, turbulence, coherent vortex structures, Chapman-Enskog expansions.